

Supplementary Material to

Measuring Economic Risk in Africa: The African Economic Risk Index and African Risk Score

September 2026

Abstract

This supplement provides material that supports the measurement and empirical results reported in the paper without interrupting its main argument. It records the domain definitions used for the eight risk pillars, the PCA conversion used for the bounded composite measures, detailed PVAR estimation and inference, expanded exact-text deduplication results, additional external-identification results, and country-level AERI and ARS outputs. The supplement uses the same analytical vintage and definitions as the paper.

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1 Detailed Pillar Descriptions

The main paper condenses the eight pillar descriptions. The definitions below record the domain interpretation used in the analytical vintage. Pillars are non-exclusive: one article can contribute to several domains when its economic content spans them.

1.1 Inflation Risk

Inflation-related reporting covers price pressures and their implications for households and firms. Its mean pillar index is 4.34 points across eligible country-months. Sustained increases can reflect greater attention to input costs, purchasing power, administered prices or anticipated price adjustment. The positive and persistent response of headline inflation to an inflation-risk innovation in the paper provides an external check on this interpretation.

1.2 FX and External Risk

External-risk reporting covers exchange-rate conditions, foreign-currency availability and external financing. Its mean pillar index is 5.03 points. Currency conditions affect imported-input costs and access to external finance, so overlap with inflation and financial-sector reporting is economically natural. The reverse-response evidence in the paper also shows that realised depreciation is followed by a higher aggregate AERI reading.

1.3 Debt and Fiscal Risk

Fiscal reporting covers public borrowing, debt service, revenue pressure and constraints on government finances. Its mean pillar index is 8.64 points. Fiscal concerns often coincide with financial-sector pressure when public financing conditions affect domestic intermediaries and with external risk when foreign-currency obligations become more salient. The overlapping classification preserves those connections within the news record.

1.4 Financial-Sector Risk

Financial-sector reporting covers intermediation, liquidity, balance-sheet pressure and financial stability. Its mean pillar index is 11.46 points, the largest pooled pillar mean in this vintage. The level describes the prominence of financial-sector concerns in the retained corpus and is interpreted together with within-pillar changes over time.

1.5 Growth and Real-Economy Risk

Real-economy reporting covers production, employment, demand and broader economic activity. Its mean pillar index is 3.17 points. News can register weakening demand or production constraints before a complete statistical account becomes available and can also respond to realised changes in activity. The reverse-response analysis in the paper shows how the aggregate AERI responds to a monthly GDP-growth innovation.

1.6 Commodity and Supply Risk

Commodity and supply reporting covers resource production, logistics and disruptions to supply. Its mean pillar index is 0.09 points. The low pooled mean reflects the concentration of such reporting in particular countries and episodes. For interpretation, changes within exposed economies are more informative than the continental average alone.

1.7 Political-Economy Risk

Political-economy reporting covers governance, policy implementation and their economic consequences. Its mean pillar index is 4.94 points. Political content enters the framework when it carries economic relevance and scored risk content, allowing governance and implementation concerns to be identified within broader high-risk episodes.

1.8 Climate and Disaster Risk

Climate and disaster reporting covers environmental disruptions and their economic effects. Its mean pillar index is 1.25 points. Such events are geographically concentrated and irregular, so the pillar is most useful as a domain-specific diagnostic read alongside country context and event data.

2 Formal PCA Conversion and Aggregation Details

Let X_j denote a bounded input, with calibration mean μ_j , standard deviation s_j and oriented first-component loading v_j . For standardized inputs $Z_j = (X_j - \mu_j)/s_j$,

$$v'Z = \sum_j \frac{v_j}{s_j} X_j - \sum_j \frac{v_j \mu_j}{s_j}. \quad (\text{S1})$$

Define $D = \sum_j v_j/s_j > 0$ and $w_j = (v_j/s_j)/D$. Then

$$100 \sum_j w_j X_j = \frac{100}{D} \left(v'Z + \sum_j \frac{v_j \mu_j}{s_j} \right). \quad (\text{S2})$$

The bounded-input score is therefore an affine rescaling of the first principal component. The implementation rejects calibration standard deviations below 10^{-10} and materially negative oriented loadings below -10^{-9} ; negligible numerical negatives are set to zero before normalization. Calculations use the unrounded weights reported in the project outputs.

Annual AERI and raw pillar indices weight monthly values by retained article counts so that annual numerators and denominators correspond to the same article universe. Annual breadth is the ratio of summed relevant articles to summed retained articles, while annual severity weights monthly conditional severity by relevant-article counts. When annual ARS or the eight-pillar PCA composite is required, the nonlinear composite is recomputed from annual inputs rather than formed as an average of monthly composite scores.

3 Detailed PVAR Specification and Inference

3.1 Macroeconomic Data and Pooled Panel Estimation

The empirical assessment relates monthly news-based risk to headline inflation, exchange-rate depreciation and monthly GDP growth. Its purpose is to test whether the measured domains display economically interpretable dynamics. ARS calibration is completed independently of these macroeconomic outcomes, preserving a clean separation between index construction and subsequent validation.

3.1.1 Monthly variables and sample construction

Headline inflation, π_{ct} , is the supplied monthly observation of the inflation rate. The exchange rate, e_{ct} , is expressed as local-currency units per US dollar. Monthly GDP, G_{ct} , is the supplied purchasing-power-parity GDP series. Depreciation and monthly GDP growth are defined as

$$d_{ct} = 100[\log e_{ct} - \log e_{c,t-1}], \quad g_{ct} = 100[\log G_{ct} - \log G_{c,t-1}]. \quad (\text{S3})$$

Thus a positive d_{ct} denotes depreciation, whereas a positive g_{ct} denotes monthly GDP growth. Headline inflation retains the definition supplied in the macroeconomic panel, while depreciation and GDP growth are the log-difference transformations above. The sample-specific winsorisation bounds are reported below.

The main validation specification uses one news-based risk measure at a time—the inflation-risk pillar, the FX and external-risk pillar, AERI or ARS—together with inflation, depreciation and monthly GDP growth. The estimation sample is distinct from the descriptive sample defined by the 100-article threshold. It begins with the 36-country union selected by the prior correlation screen, then retains country-months with the risk measure, the required macroeconomic inputs and the full lag structure, subject to at least 24 usable observations per country.¹ The common GDP-augmented panel contains 4,286 observations for 36 countries and runs from May 2015 to December 2025. The 100-article rule continues to govern the principal descriptive country comparisons and ARS calibration; it is not imposed on the PVAR estimation panel. This distinction explains why a country can appear in the PVAR sample even when few or no months satisfy the descriptive coverage threshold. Using the direct pillar indices in this specification keeps the economic interpretation aligned with the measurement framework in the main text. Supporting transformations and country-level estimation diagnostics are available from the authors on request.

Risk innovations are put on a comparable within-country scale. Let $\bar{r}_c$ be the country mean in the supplied scaling panel and $s_{r,w}$ the sample standard deviation of $r_{ct} - \bar{r}_c$ pooled across that panel. We use $\tilde{r}_{ct} = r_{ct}/s_{r,w}$; country effects subsequently absorb level differences. A one-unit risk innovation therefore represents one pooled within-country standard deviation, rather than one raw index point. Scaling parameters are fixed for the reported estimates and bootstrap exercises.

¹The estimation panel winsorises the supplied macroeconomic transformations at pooled first and ninety-ninth percentiles. In the GDP-augmented sample the retained bounds are $[-2.565, 50.934]$ for inflation, $[-5.736, 14.926]$ for depreciation and $[-0.819, 1.506]$ for monthly GDP growth.

3.1.2 The pooled dynamic system

For m endogenous variables, the maintained model is

$$y_{ct} = \alpha_c + \lambda_t + \sum_{\ell=1}^p B_{\ell} y_{c,t-\ell} + \varepsilon_{ct}, \quad E(\varepsilon_{ct} \varepsilon'_{ct}) = \Sigma. \quad (\text{S4})$$

Here α_c captures persistent country differences, and λ_t is a full year-month effect that absorbs shocks common to countries in a given month. The year-month effects therefore absorb common calendar-time variation rather than recurring month-of-year seasonality alone. The slope matrices B_{ℓ} are pooled across countries. In the three-variable system, $y_{ct} = (\tilde{r}_{ct}, \pi_{ct}, d_{ct})'$. In the GDP-augmented system, $y_{ct} = (\tilde{r}_{ct}, \pi_{ct}, d_{ct}, g_{ct})'$. In particular, the inflation equation in the latter system is

$$\begin{aligned} \pi_{ct} = \alpha_c^{\pi} + \lambda_t^{\pi} + \sum_{\ell=1}^p (b_{\pi r, \ell} \tilde{r}_{c,t-\ell} + b_{\pi \pi, \ell} \pi_{c,t-\ell} \\ + b_{\pi d, \ell} d_{c,t-\ell} + b_{\pi g, \ell} g_{c,t-\ell}) + \varepsilon_{ct}^{\pi}. \end{aligned} \quad (\text{S5})$$

The risk, depreciation and GDP equations have the same lag structure with equation-specific coefficients. Each outcome can therefore respond to its own history and to the histories of the other variables. Pooling provides a common dynamic description conditional on fixed effects. The multivariate panel structure follows the empirical framework in Love and Zicchino (2006) and Abrigo and Love (2016), implemented here with within-transformed pooled least squares.

Let X stack all mp lagged regressors and Y stack the contemporaneous outcomes on the common estimation sample. Alternating country and year-month projections yield $\ddot{X}$ and $\ddot{Y}$. The pooled estimator and residual covariance are

$$\hat{B} = (\ddot{X}' \ddot{X})^+ \ddot{X}' \ddot{Y}, \quad \hat{E} = \ddot{Y} - \ddot{X} \hat{B}, \quad \hat{\Sigma} = \frac{\hat{E}' \hat{E}}{n - mp}. \quad (\text{S6})$$

The superscript $+$ denotes the Moore–Penrose inverse; the covariance denominator is the implemented lag-regressor adjustment. A common maximum-lag sample is used when comparing candidate orders, preventing changes in the available observations from driving lag choice. Among stable candidates with $p \in \{1, 2, 3\}$, the criterion is

$$\text{BIC}(p) = \log |\hat{\Sigma}_p| + \frac{\log n}{n} m^2 p. \quad (\text{S7})$$

Stability requires the maximum absolute companion-matrix eigenvalue to lie below the numerical threshold of 0.999. All reported specifications select three lags. Table S1 reports observations, country counts, roots and accepted bootstrap draws. Fixed-effects estimation with lagged outcomes can retain finite-time bias (Nickell, 1981); the long monthly dimension reduces some finite-sample concerns but leaves this source of bias relevant to interpretation.

3.1.3 Identification and impulse responses

Let $\hat{\Sigma} = LL'$ with L lower triangular. Forward systems order risk first, followed by inflation, depreciation and monthly GDP growth where included. This permits a risk innovation to affect the macroeconomic variables within the month, while assigning contemporaneous feedback into risk to subsequent periods. For reverse responses, the order is inflation, depreciation, monthly GDP growth and AERI. The two orderings answer complementary recursive questions about forward risk propagation and macroeconomic feedback.

The moving-average matrices and normalised responses are

$$\Phi_0 = I_m, \quad \Phi_h = \sum_{\ell=1}^{\min(p,h)} \hat{B}_\ell \Phi_{h-\ell}, \quad (S8)$$

$$\theta_{kj}(h) = \frac{e'_k \Phi_h L e_j}{e'_j L e_j}, \quad h = 0, \dots, 12, \quad (S9)$$

where e_j is the j th coordinate vector, distinct from the exchange-rate notation in equation (S3). The denominator normalises the innovation to a one-unit contemporaneous increase in the selected variable. For forward risk innovations, that unit is one within-country standard deviation. For reverse macroeconomic innovations, it is one percentage point. This convention makes the reported response magnitudes explicit while preserving the natural units used for the macroeconomic innovations.

3.1.4 Inference and interpretation

Uncertainty bands are constructed by resampling countries with replacement, retaining each selected country's complete time history, and re-estimating the fixed-effects system. The lag order, measurement transformations and fixed ARS calibration remain unchanged across bootstrap replications. Four hundred accepted stable replications provide the fifth and ninety-fifth percentiles at each horizon. These are pointwise 90 percent intervals, so each band describes uncertainty at a given horizon. Simultaneous inference for the full response path or for the location of its population peak would require a different procedure.

Joint lag tests complement the IRFs. For example, the inflation equation tests

$$H_0 : b_{\pi r,1} = \dots = b_{\pi r,p} = 0, \quad W = (R\hat{\beta})'[R\hat{V}R']^+(R\hat{\beta}), \quad (S10)$$

using country-cluster covariance $\hat{V}$ and a χ_p^2 reference distribution. The exact covariance calculation is given in Appendix 3. Rejection indicates incremental information in the included lagged variable conditional on the other lags and fixed effects. The recursive response additionally reflects contemporaneous residual covariance and propagation through the full system, so the tables report both forms of evidence.

The 36-country set is the inclusive union of countries whose earlier country-level screening exercise produced an absolute correlation of at least 0.30 for either inflation-risk versus inflation or FX/external-risk versus the exchange rate. Twenty-eight countries enter through the inflation screen, fourteen through the exchange-rate screen and six through both. The set comprises Algeria, Angola, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Comoros, Democratic

Table S1: Pooled PVAR samples and stability diagnostics

System	Variables	Countries	Obs.	Lags	Max. root	Draws
Inflation risk + GDP	4	36	4,286	3	0.9228	400/400
FX/external risk + GDP	4	36	4,286	3	0.9222	400/400
AERI + GDP	4	36	4,286	3	0.9198	400/400
ARS + GDP	4	36	4,286	3	0.9190	400/400

Notes: All samples span May 2015–December 2025 and use the same 36-country correlation-selected panel. The PVAR sample is formed from complete model inputs and lags with at least 24 usable observations per country; the 100-article rule used for descriptive country comparisons is not imposed here. Draws report accepted/attempted country-cluster replications. A companion root below one indicates fitted dynamic stability; identification is assessed separately through the stated recursive and external-shock designs.

Republic of the Congo, Egypt, Ethiopia, Gambia, Ghana, Guinea, Guinea-Bissau, Ivory Coast, Lesotho, Liberia, Libya, Malawi, Mali, Mauritius, Morocco, Mozambique, Namibia, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Somalia, South Africa, Tanzania, Tunisia, Uganda, Zambia and Zimbabwe.

The PVAR estimation sample should be read separately from the descriptive coverage sample reported later in this supplement. AERI and the pillar series are available for lower-volume country-months, whereas the 100-article threshold is used to define the principal descriptive comparisons and the ARS calibration sample. The PVAR retains the correlation-selected countries with complete model inputs and the required lag structure and imposes a minimum of 24 usable observations per country. This is why Guinea-Bissau and Sao Tome and Principe can enter the PVAR even though Table S7 reports zero and twenty months, respectively, under the stricter 100-article descriptive rule.

For equation k , write the country-cluster score as $s_{ck} = \ddot{X}'_c \hat{\varepsilon}_{ck}$, let $q = mp$, and denote the number of countries by C . The covariance used in the joint lag tests is

$$\hat{V}_k = \frac{C}{C-1} \frac{n-1}{n-q} (\ddot{X}' \ddot{X})^+ \left(\sum_{c=1}^C s_{ck} s'_{ck} \right) (\ddot{X}' \ddot{X})^+. \quad (\text{S11})$$

Country resampling preserves dependence within each country’s observed history. Year-month effects absorb common calendar-time variation. Inference is conditional on the correlation-screened country set and on the fixed ARS calibration.

4 Exact-Text Deduplication: Expanded Results

Repeated publication can alter a country-month index when identical high-scoring text appears in several records. The robustness exercise decodes article bodies, restricts comparison to bodies of at least 200 characters, and collapses exact matches within each country-month. Appearances in different countries or different months remain separate observations. Edited stories, translations and rewritten versions also remain separate because the test is deliberately exact.

The reconstruction covers 21,303,020 frozen-vintage records in 7,126 country-months. It is 553 records and one Togo country-month short of the archived aggregate because 42 source files were empty or unreadable. Where a reconstructed month does not fully match the archived

count, unmatched frozen-vintage records remain in the denominator and only directly observed duplicates are removed. This preserves the archived universe while allowing the contribution of verified repetitions to be measured.

Table S2: Sensitivity to within-country-month exact-text deduplication

Statistic	Result
Records removed	497,091
Removed share (percent)	2.33
Cross-outlet copies removed	186,379
AERI Pearson correlation	0.9296
AERI rank correlation	0.9883
AERI mean absolute change	2.32
AERI 95th-percentile absolute change	8.30
ARS Pearson correlation	0.9680
ARS mean absolute change	0.61
Months falling below 100 articles	69

Notes: The comparison uses 6,418 country-months with at least 100 articles before and after deduplication. Exact duplicates have the same decoded body within a country-month; bodies shorter than 200 characters are retained. Both numerator and denominator are adjusted. ARS uses the original unrounded PCA weights.

Figure S1 compares the monthly cross-country median AERI before and after exact-text removal on the common coverage sample. The two paths retain the same broad chronology and both peak in April 2020. The differences are concentrated in particular countries and months rather than spread evenly through the panel.

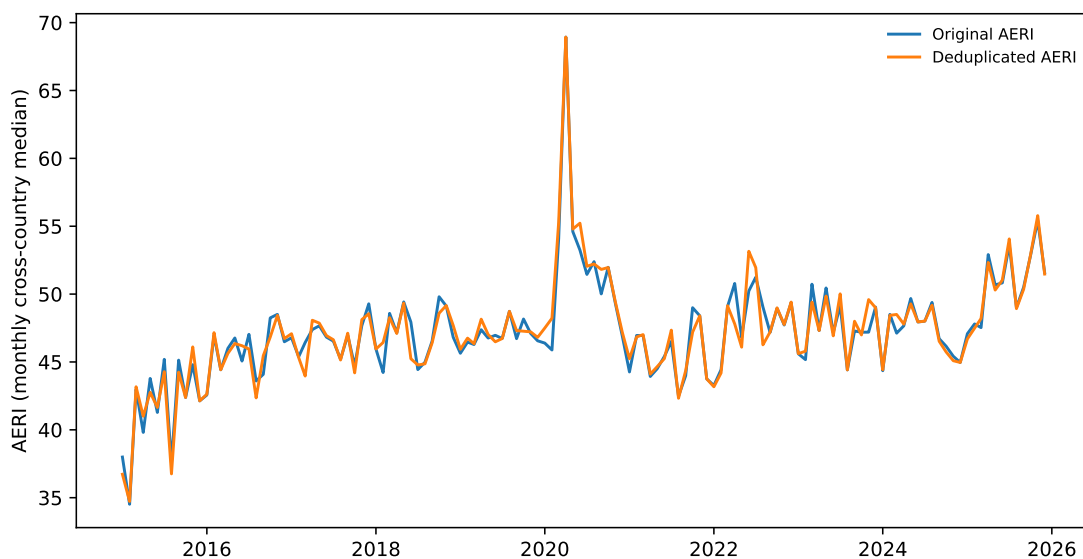


Figure S1: Monthly median AERI before and after within-country-month exact-text deduplication

Notes: Each line is the cross-country median among country-months in the common coverage sample used for the deduplication comparison. The aggregate peak remains April 2020.

The median within-country AERI correlation is 0.9992 and 48 of 53 countries retain the same peak month on the common sample. Sierra Leone is the largest exception: its November 2016 observation falls from 269.72 to 110.77 and the deduplicated peak moves to October 2016. Rwanda's August 2021 observation falls from 95.43 to 63.35, while Mauritius rises from 91.64 to 111.79 in May 2020 because removal changes both the numerator and denominator. These cases

Table S3: Country-level sensitivity of AERI peaks to exact-text deduplication

Country	Original	Dedup. at original	Δ	Within-country r	Dedup. peak
Sierra Leone	269.72	110.77	-158.95	0.812	2016-10
Rwanda	95.43	63.35	-32.08	0.984	2020-04
Mauritius	91.64	111.79	20.15	0.998	2020-05
Angola	85.67	72.69	-12.98	0.932	2020-04
Eswatini	75.11	87.01	11.89	0.908	2020-04
Zambia	121.49	130.62	9.12	0.684	2024-09
Cameroon	68.83	60.92	-7.91	0.992	2016-12
Botswana	90.44	85.72	-4.72	0.972	2025-12

Notes: Countries shown are those with the largest absolute change at the original peak, augmented to retain Rwanda and Mauritius because their peak behavior is discussed in the paper. The correlation is computed over the common within-country monthly sample. A change can be positive because removing duplicate records changes both the score numerator and the all-article denominator.

identify where syndicated or repeated identical text materially affects a country history while leaving the continental chronology largely unchanged.

5 External Monetary-Policy Exposure: Additional Results

The external-identification exercise uses the pure U.S. monetary-policy surprise from Jarociński and Karadi (2020), aggregated to the month and standardized over the 2015–2025 sample. Predetermined exposure is the 2014 stock of external debt as a share of GNI from the World Bank, transformed as $\log(1 + x)$ and standardized. For country i and horizon h , the preferred local projection is

$$\begin{aligned} \text{AERI}_{i,t+h} - \text{AERI}_{i,t-1} = & \alpha_i^h + \gamma_t^h + \beta_h(MP_t \times E_i^{2014}) \\ & + \sum_{\ell=1}^3 \theta_{h\ell} \Delta \text{AERI}_{i,t-\ell} + \sum_{\ell=1}^3 \psi_{h\ell} (MP_{t-\ell} \times E_i^{2014}) + \varepsilon_{i,t+h}. \end{aligned} \quad (\text{S12})$$

Country effects absorb persistent differences in average AERI and year-month effects absorb shocks shared by the African sample in a given month. The identifying variation is the cross-country difference in predetermined exposure to the same externally measured tightening. Standard errors are two-way clustered by country and month, and finite-cluster inference uses 28 denominator degrees of freedom for the 29-country cross-section.

The horizon path is delayed. The differential response reaches 0.272 AERI points at month four, with a 90 percent interval of $[0.108, 0.436]$, and 0.384 points at month nine, with interval $[0.085, 0.684]$. The six reported robustness variants retain positive month-four and month-nine coefficients. Leave-one-country-out estimates remain positive across all 29 omissions. Pre-shock outcomes and the central-bank-information component of the same announcements do not reproduce the exposure pattern, while the finite-cluster and restricted wild-cluster references place the joint future-shock placebo above conventional significance thresholds. Complete monthly shock, exposure and intermediate estimation files are available from the authors on request.

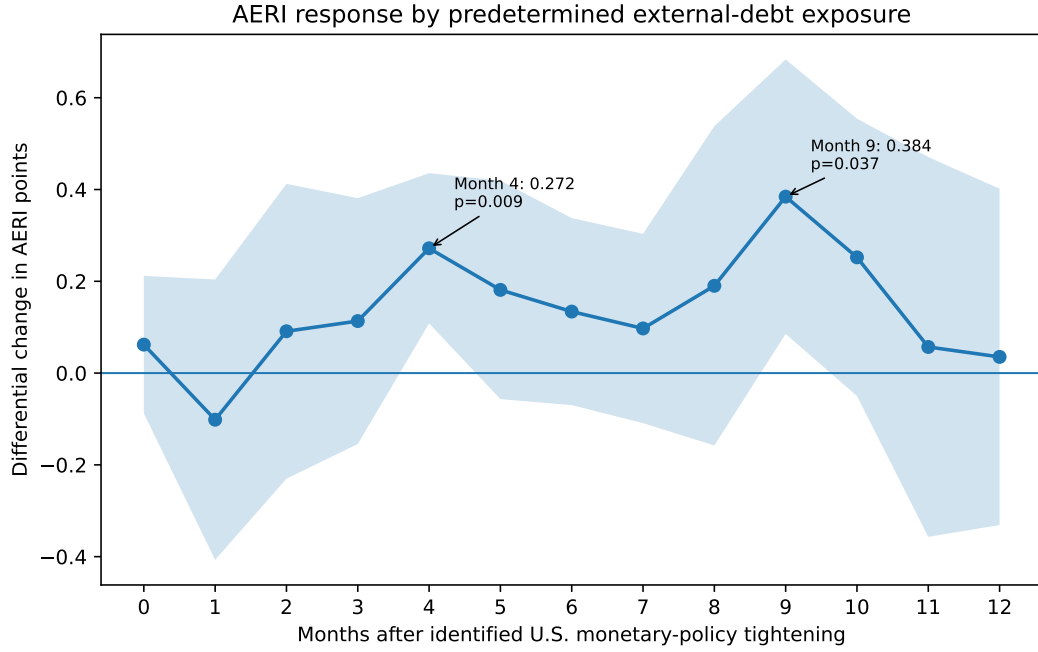


Figure S2: Differential AERI response to an externally identified U.S. monetary-policy tightening
Notes: The line reports $\hat{\beta}_h$. Shading gives pointwise 90 percent finite-cluster intervals. Exposure is standardized $\log(1 + 2014 \text{ external debt/GNI})$.

Table S4: Full horizon path for the external monetary-policy exposure specification

Horizon	$\hat{\beta}_h$	90% CI lower	90% CI upper	<i>p</i> -value
0	0.062	-0.088	0.212	0.487
1	-0.102	-0.407	0.204	0.576
2	0.091	-0.230	0.412	0.633
3	0.113	-0.155	0.381	0.478
4	0.272	0.108	0.436	0.009
5	0.181	-0.057	0.419	0.206
6	0.134	-0.070	0.338	0.273
7	0.097	-0.109	0.304	0.430
8	0.190	-0.158	0.538	0.360
9	0.384	0.085	0.684	0.037
10	0.252	-0.050	0.554	0.167
11	0.057	-0.357	0.471	0.817
12	0.035	-0.331	0.402	0.871

Notes: The coefficient is the differential cumulative AERI response to a one-standard-deviation U.S. monetary-policy tightening for a country one standard deviation higher in predetermined 2014 external-debt exposure. Intervals and *p*-values use the finite-cluster reference with 28 denominator degrees of freedom.

Table S5: Externally identified monetary-policy exposure validation

Specification	$\hat{\beta}_4$	p_4	$\hat{\beta}_9$	p_9
Primary: log debt/GNI	0.272	0.0086	0.384	0.0373
Raw debt/GNI	0.203	0.0152	0.289	0.0556
95th-percentile-capped debt/GNI	0.230	0.0284	0.347	0.0566
Exclude Mozambique	0.265	0.0305	0.390	0.0610
African-country panel restricted to FOMC announcement months	0.299	0.0078	0.354	0.0395
Alternative monetary-policy shock	0.251	0.0068	0.420	0.0534
Leave-one-country-out range	[0.214, 0.308] 29/29 $p < 0.10$ [0.282, 0.455] 29/29 $p < 0.10$			

Notes: The dependent variable is the cumulative change in AERI from month $t - 1$ through horizon h . The shock is the standardized pure U.S. monetary-policy surprise from the Jarociński–Karadi high-frequency decomposition, aggregated to the month. Predetermined exposure is 2014 external debt stocks as a share of GNI from the World Bank, transformed as $\log(1 + x)$ and standardized in the primary specification. All regressions include country and year-month fixed effects, three lags of the AERI change and three lags of the shock–exposure interaction. Standard errors are two-way clustered by country and month. Reported p -values use a finite-cluster t reference with degrees of freedom equal to the number of African-country clusters minus one. Restricted wild-cluster-bootstrap p -values for the primary month-four and month-nine coefficients are 0.042 and 0.034, respectively. In the leave-one-country-out exercise, 28 of 29 month-four estimates and 24 of 29 month-nine estimates also satisfy $p < 0.05$.

Table S6: External-identification diagnostics and falsification checks

Check	Statistic / range	Interpretation
Pre-shock AERI-change placebos (1–6 months)	$p \in [0.156, 0.807]$	None significant at 10 percent
CBI shock $\times$ exposure, month 4	$p = 0.153$	p-value above 0.10
CBI shock $\times$ exposure, month 9	$p = 0.125$	p-value above 0.10
Joint future-shock interaction placebo	$\chi^2(6) = 11.30$, asymptotic $p = 0.079$	Large-sample reference only
Finite-cluster joint reference	$F(6, 28) = 1.884$, $p = 0.119$	p-value above 0.10
Restricted wild-cluster bootstrap	$p = 0.221$	p-value above 0.10

Notes: The joint future-shock placebo includes the contemporaneous monetary-policy-shock–exposure interaction as a control. The asymptotic Wald reference is reported for transparency, but the preferred joint inference recognizes that the identifying cross-section contains 29 African-country clusters. The finite-cluster F reference uses 28 denominator degrees of freedom; the restricted wild-cluster bootstrap uses 999 Rademacher replications by African country. The pre-shock outcome placebos are estimated without lagged dependent-variable controls that would mechanically span the placebo outcome.

6 All-Country AERI and ARS Outputs

Table [S7](#) reports compact AERI and ARS outputs for all 54 economies in the 2015–2025 panel. The month count refers only to country-months with at least 100 retained articles. It is therefore a descriptive-coverage statistic rather than a PVAR eligibility count.

Table S7: Country-level AERI and ARS summary statistics, 2015–2025

Country	Months	Mean AERI	Max AERI	AERI peak	Mean ARS	Max ARS	ARS peak
Algeria	132	65.02	83.44	2020-04	44.67	50.57	2020-04
Angola	132	53.45	85.67	2019-10	39.45	49.07	2020-05
Benin	132	43.38	71.98	2020-04	36.28	45.65	2020-04
Botswana	132	69.39	90.44	2025-12	43.67	49.59	2025-12
Burkina Faso	132	47.73	68.40	2020-04	37.90	46.83	2020-04
Burundi	132	49.20	84.80	2022-08	37.84	49.40	2022-08
Cameroon	132	48.06	68.83	2018-11	38.77	46.58	2018-11
Cape Verde	131	39.73	67.21	2022-07	31.79	42.38	2022-06
Central African Republic	129	62.25	116.38	2017-04	39.97	53.18	2017-04
Chad	96	55.64	103.13	2020-04	38.72	54.16	2020-04
Comoros	130	42.83	73.75	2022-06	36.58	47.39	2025-02
Republic of the Congo	132	61.63	87.31	2020-05	41.42	48.92	2020-05
Democratic Republic of the Congo	132	55.00	85.25	2020-08	38.92	48.42	2022-04
Djibouti	4	64.96	73.62	2024-05	41.17	45.06	2024-05
Egypt	132	40.55	97.11	2025-12	35.94	51.26	2025-12
Equatorial Guinea	73	21.10	45.64	2020-05	18.78	28.19	2020-05
Eritrea	132	49.29	77.55	2024-06	37.40	46.61	2021-03
Eswatini	131	39.94	75.11	2020-04	33.48	45.43	2020-04
Ethiopia	132	59.96	127.83	2019-02	41.11	60.87	2019-02
Gabon	132	54.49	300.16	2016-05	36.82	85.37	2016-05
Gambia	132	58.36	100.99	2020-04	40.33	52.48	2020-04
Ghana	132	57.11	85.70	2025-03	40.84	49.75	2025-03
Guinea	132	41.24	91.32	2018-07	35.85	50.04	2018-07

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Table S7 – continued from previous page

Country	Months	Mean AERI	Max AERI	AERI peak	Mean ARS	Max ARS	ARS peak
Guinea-Bissau	0	–	–	–	–	–	–
Ivory Coast	132	30.22	46.87	2025-04	31.23	37.87	2020-04
Kenya	132	38.85	71.03	2025-12	35.01	44.12	2025-12
Lesotho	132	74.17	111.28	2020-04	44.79	54.89	2025-06
Liberia	118	90.79	128.07	2020-04	49.41	58.71	2016-08
Libya	132	12.77	32.94	2024-09	27.35	37.15	2024-09
Madagascar	132	23.83	53.24	2020-06	30.10	41.60	2020-06
Malawi	132	56.18	93.32	2025-02	40.11	51.12	2025-02
Mali	132	61.90	82.69	2016-09	43.35	49.62	2016-09
Mauritania	132	1.00	30.51	2019-06	2.26	31.32	2019-06
Mauritius	132	19.05	91.64	2020-05	29.70	52.02	2020-05
Morocco	132	22.67	37.66	2025-10	31.17	36.68	2020-04
Mozambique	132	34.16	61.17	2020-04	33.05	41.84	2020-04
Namibia	132	49.96	73.27	2025-12	37.97	44.78	2025-12
Niger	132	28.72	51.58	2025-12	29.69	38.50	2020-04
Nigeria	132	73.04	98.43	2024-02	46.21	52.76	2024-02
Rwanda	110	39.31	95.43	2021-08	32.79	51.79	2021-08
Sao Tome and Principe	20	33.90	44.72	2020-07	24.94	32.31	2021-12
Senegal	132	25.47	49.80	2025-02	32.15	40.24	2025-02
Seychelles	2	10.98	11.43	2025-06	20.94	21.07	2025-06
Sierra Leone	132	131.61	269.72	2016-11	58.50	79.56	2016-11
Somalia	132	8.26	48.54	2025-11	24.59	40.23	2025-11
South Africa	132	42.58	79.22	2020-04	36.71	48.74	2020-04
South Sudan	132	48.74	98.09	2025-07	37.21	51.08	2025-07

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Table S7 – continued from previous page

Country	Months	Mean AERI	Max AERI	AERI peak	Mean ARS	Max ARS	ARS peak
Sudan	132	5.05	17.15	2024-07	28.03	36.93	2019-02
Tanzania	132	14.88	54.45	2025-12	20.39	39.41	2025-12
Togo	132	72.05	108.08	2020-04	43.70	54.77	2020-04
Tunisia	132	47.51	74.22	2022-03	38.36	47.17	2022-03
Uganda	132	60.50	84.86	2025-07	41.33	47.93	2025-07
Zambia	132	96.43	121.49	2024-09	50.71	57.41	2025-06
Zimbabwe	132	64.99	98.27	2019-01	42.55	51.11	2019-01

Notes: Months count country-months with at least 100 retained articles. AERI is the headline article-score index. ARS is the fixed PCA-weighted 0–100 African Risk Score used in the main analysis. Peak months are identified within January 2015–December 2025. Guinea-Bissau has no month meeting the 100-article descriptive threshold in this vintage and is therefore shown with missing summary statistics rather than a misleading low-risk value. These month counts describe the 100-article descriptive sample and do not determine inclusion in the PVAR, which uses a separate correlation-screened, macro-data-complete estimation sample. The complete monthly panel is available from the authors on request.

7 Individual Country AERI Dynamics

This section reports monthly AERI for the ten economies used in the annual comparison in the paper: Nigeria, Ghana, Algeria, Morocco, Egypt, Ethiopia, South Africa, Kenya, Angola and Ivory Coast. The dashed reference line in the figures is the contemporaneous median across eligible countries.

Table S8: Country-level AERI: descriptive comparisons, 2015–2025

Country	Months	Mean	SD	Peak	Peak month
Nigeria	132	73.04	8.75	98.43	2024-02
Ghana	132	57.11	10.47	85.70	2025-03
Algeria	132	65.02	6.34	83.44	2020-04
Morocco	132	22.67	7.54	37.66	2025-10
Egypt	132	40.55	11.29	97.11	2025-12
Ethiopia	132	59.96	17.50	127.83	2019-02
South Africa	132	42.58	7.61	79.22	2020-04
Kenya	132	38.85	7.02	71.03	2025-12
Angola	132	53.45	8.60	85.67	2019-10
Ivory Coast	132	30.22	5.28	46.87	2025-04

Notes: Monthly AERI uses the article-score definition reported in the paper. Only months with at least 100 retained articles are included. Means and sample standard deviations give equal weight to eligible months; they differ from article-weighted annual AERI. Peak denotes the maximum eligible monthly observation, with the earliest month reported in a tie. These ten economies form the illustrative group used in the paper; PVAR sample selection is conducted separately.

The country series combine common shocks with marked differences in levels, volatility and timing. Nigeria remains high through much of the sample and peaks in February 2024. Ethiopia has the largest standard deviation in the ten-country group and records its maximum in February 2019. South Africa and Algeria show pronounced April 2020 peaks, while Morocco’s series rises more gradually later in the sample. The common vertical scale preserves these differences rather than normalizing each country to its own history.



Figure S3: Monthly country AERI: Nigeria, Ghana, Algeria, Morocco, Egypt and Ethiopia
Notes: Solid lines show monthly AERI. The dashed line is the median across countries with at least 100 retained articles in the corresponding month. No smoothing or country-specific rebasing is applied.

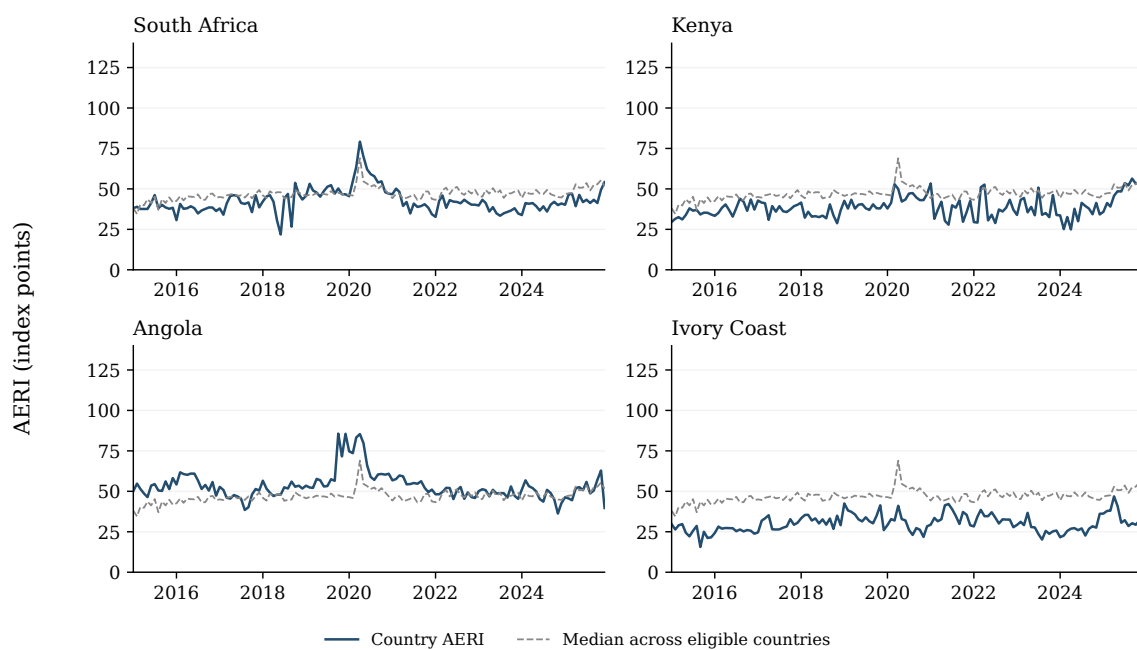


Figure S4: Monthly country AERI: South Africa, Kenya, Angola and Ivory Coast
Notes: Definitions, timeline and vertical scale match the preceding figure. The reference median varies with the composition of eligible countries.

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